

Foundation of Research

BCAC702 Research Methodology

Presented by **Dr. Tumpa Banerjee**



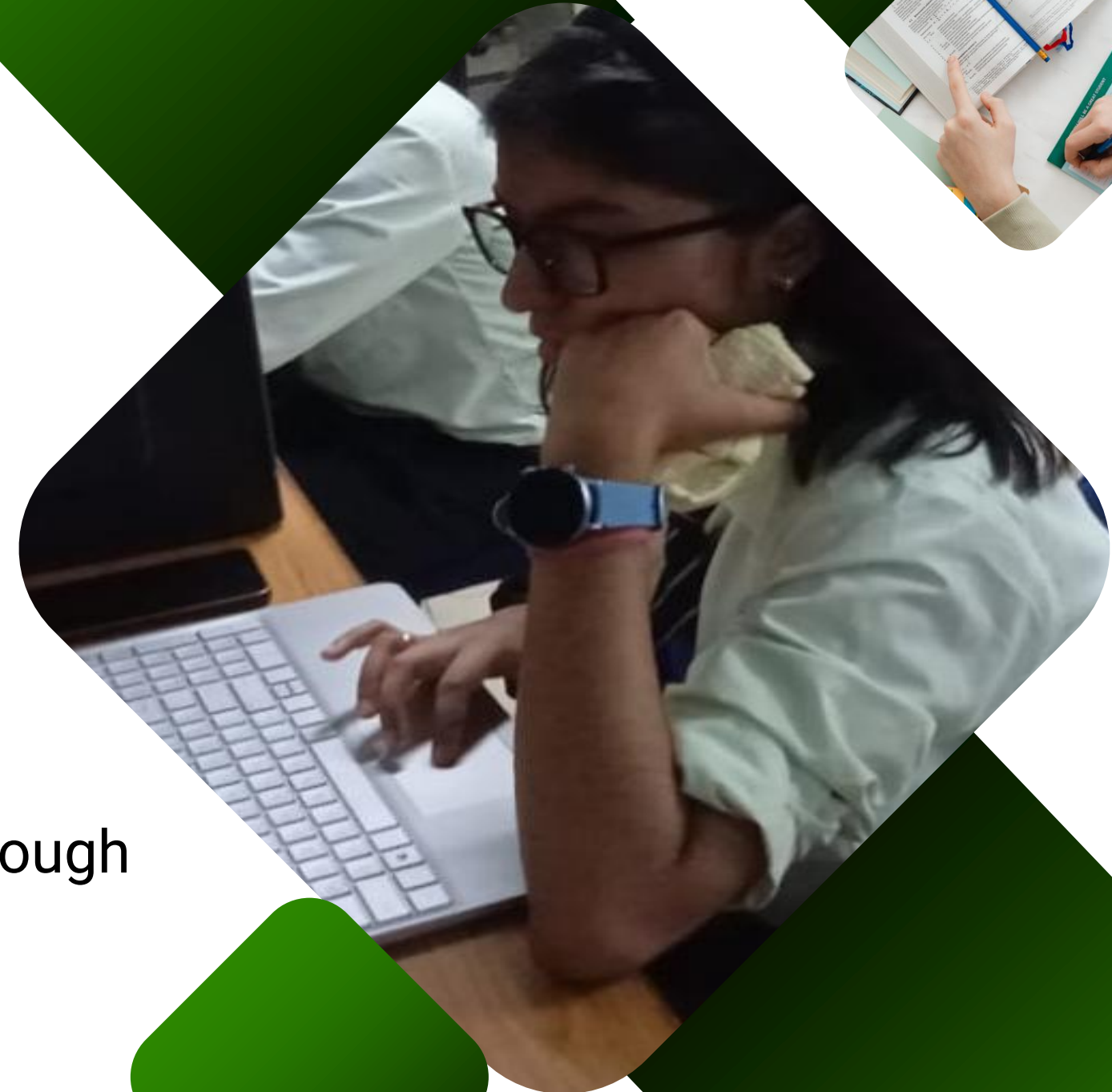
www.sittechno.edu.in



What is Research?

AN ART OF SCIENTIFIC INVESTIGATION

- A careful investigation or inquiry specially through search for new facts in any branch of knowledge.
- Systematized effort to gain new knowledge



What is Research?

AN ART OF SCIENTIFIC INVESTIGATION

According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organising and evaluating data; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis.

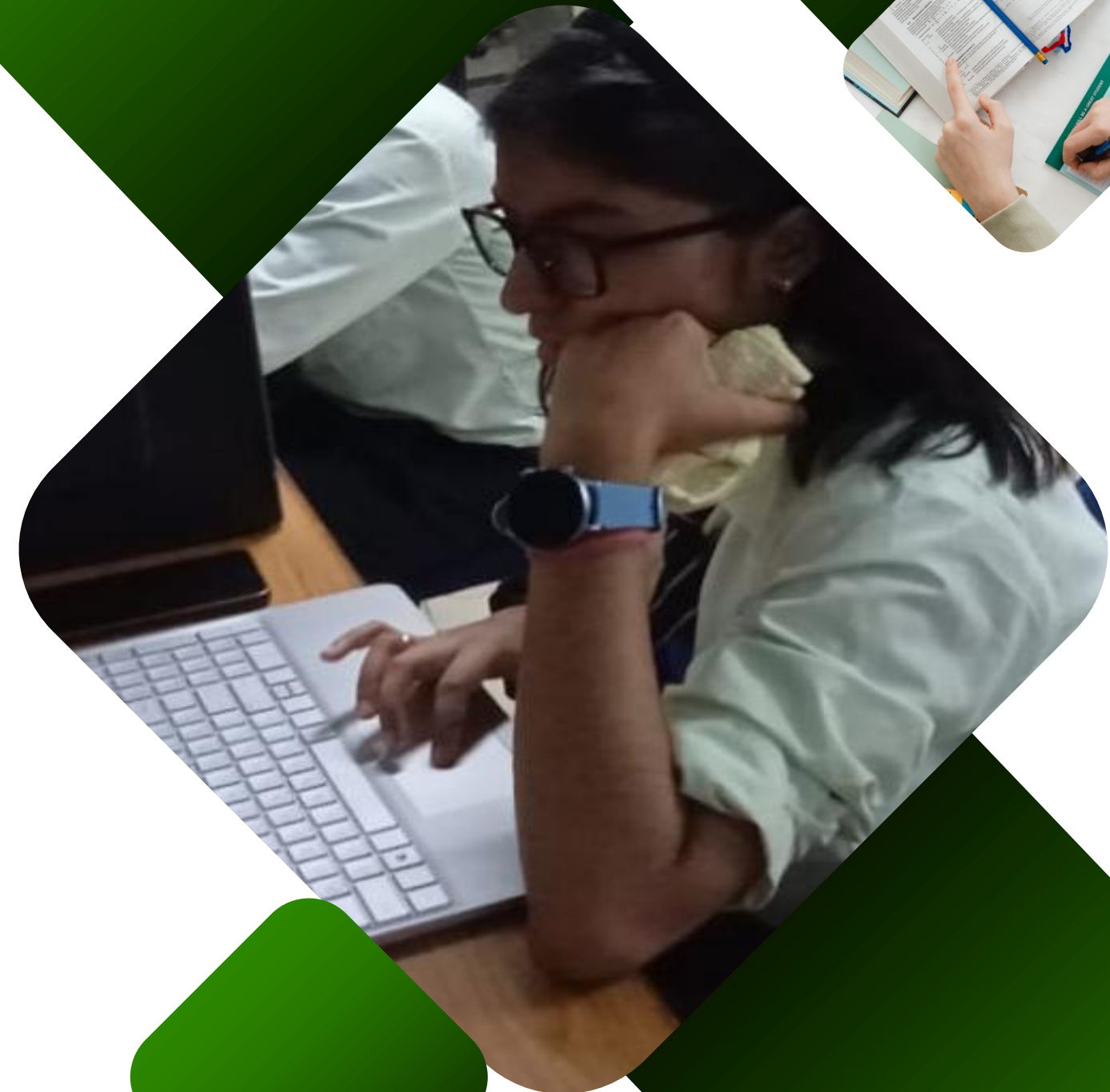


What is Research?

AN ART OF SCIENTIFIC INVESTIGATION

D. Slesinger and M. Stephenson in the Encyclopaedia of Social Sciences define research as “the manipulation of things, concepts or symbols for the purpose of generalising to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art.”

Research is, thus, an original contribution to the existing stock of knowledge making for its advancement. It is the pursuit of truth with the help of study, observation, comparison and experiment.



Activity

Ordinary Project	Convert into Research
Student attendance app	Which authentication method provides better security and usability for attendance applications?
E-commerce website	How does recommendation strategy affect user engagement in an e-commerce system?
Chatbot	How do different prompting strategies affect the accuracy of an LLM chatbot?
Face recognition system	How does image quality affect face-recognition performance?
Object detection system	Which YOLO model provides the best accuracy–speed trade-off for real-time object detection?

KEY RESEARCH OBJECTIVES

EACH RESEARCH STUDY HAS ITS OWN SPECIFIC PURPOSE

OBJECTIVE 1: UNDERSTAND

To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as exploratory or formulative research studies);

OBJECTIVE 2: ANALYZE

To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies);

OBJECTIVE 3: PROPOSE

To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies);
To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies).

Types of Research

- Descriptive vs Analytical
- Applied vs Fundamental
- Quantitative vs Qualitative
- Conceptual vs Empirical



DESCRIPTIVE VS ANALYTICAL

In analytical research the researcher has to use facts or information already available and analyze these to make a critical evaluation of the material

- Descriptive research include surveys and fact finding enquiries of different kinds
- description of the state of art affairs as it exists at present
- The researcher has no control over the variable. He can only report what has happened or what is happening
- It is also called ex post facto research



Descriptive Research

Describes the current situation, characteristics, trends, or behavior without necessarily explaining why it happens.

- Social Media Usage: Study how much time college students spend on social media and which platforms they use most.
- Programming Language Preference: Survey BCA students to find out which programming languages they prefer—Python, Java, C++, etc.
- Cybersecurity Awareness: Study the level of cybersecurity awareness among college students.
- AI Tool Usage: Find out how many students use ChatGPT, Gemini, or other AI tools for programming.

Analytical Research

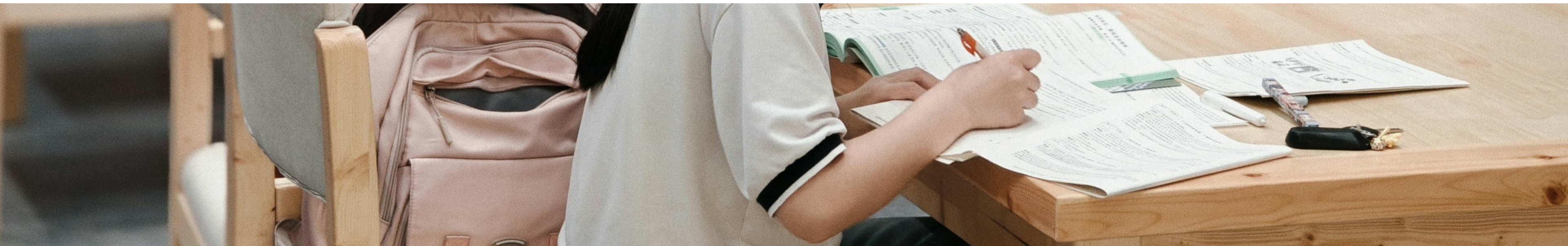
Goes beyond description and analyzes relationships, causes, differences, or patterns in the data.

- Analyze whether programming experience affects programming performance.
- Analyze whether increased use of AI coding assistants improves programming productivity.
- Compare the performance of YOLOv8 and YOLO11 for object detection.
- Analyze whether network latency affects user satisfaction in cloud applications.

APPLIED VS FUNDAMENTAL

Fundamental research is mainly concerned with the generalization and with the formulation of a theory. It is directed towards finding information that has a broad base of application and thus adds to the already existing organized body of scientific knowledge.

Applied research aims at finding a solution for an immediate problem facing a society or an industrial or business organization. Research aimed at certain conclusion facing a concrete social or business problem is an example of applied research.



Applied Research

Research conducted to solve a specific practical or real-world problem.

- Developing an AI system to detect helmet violations from CCTV footage.
- Developing a machine-learning model to detect spam emails.
- Designing a chatbot for answering college admission queries.
- Developing an application to detect plant diseases using images.
- Developing an algorithm to reduce cloud computing latency.

Fundamental Research

Research aimed at developing new knowledge or understanding without an immediate practical application.

- Investigating how neural networks learn hierarchical representations.
- Studying the theoretical properties of a new optimization algorithm.
- Investigating why certain neural network architectures generalize better than others.
- Developing a mathematical model for understanding the convergence behavior of a learning algorithm.
- Studying the theoretical limits of computational complexity.

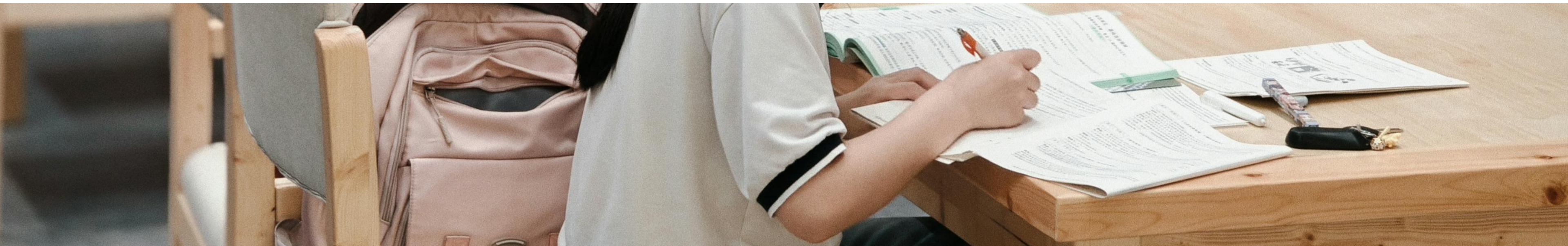
QUANTITATIVE VS QUALITATIVE

Quantitative research is based on the quantitative measurement of some characteristics

It is applicable to phenomena that can be expressed in terms of quantity

Qualitative research on the other hand concern with qualitative phenomenon

This type of research aims at discovering the underlying motive and desires using in depth interviews for the purpose



Quantitative Research

Research based primarily on numerical data, measurements, statistics and mathematical analysis.

- Compare the accuracy of three machine-learning models.
- Measure the execution time of different sorting algorithms.
- Measure network latency under different workloads.
- Compare YOLO models using Precision, Recall, F1-score and mAP.
- Measure the effect of different optimizers on neural-network accuracy.
- Measure the response time of different databases.

Qualitative Research

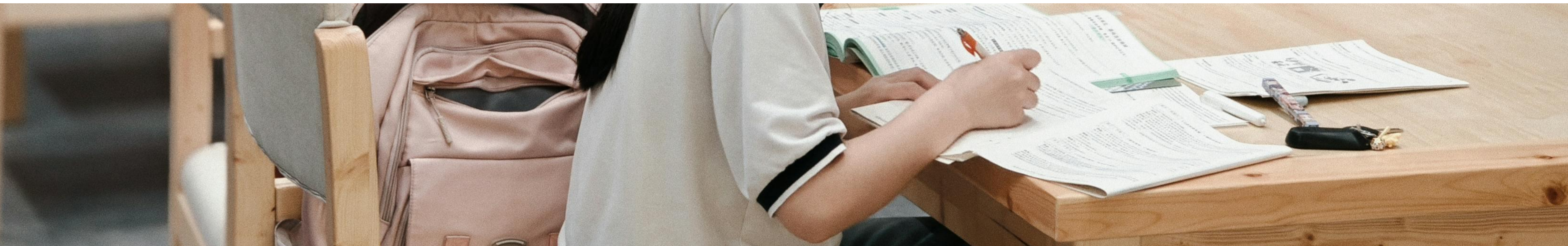
Research based primarily on opinions, experiences, perceptions, observations, interviews, textual data, etc.

- Interview students about their experiences using AI coding assistants.
- Study developers' perceptions of using ChatGPT for software development.
- Investigate users' experiences with a mobile banking application.
- Interview software developers about challenges in adopting cloud computing.
- Study students' perceptions regarding AI-generated code.

CONCEPTUAL VS EMPIRICAL

Conceptual research is related to some abstract ideas or theory
It is generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones

Empirical research relies on experience or observation alone, often without due regard of for system and theory
The researcher must first provide himself with a working hypothesis to guess as to the probable result he then works to get enough facts or data to prove or disprove that hypothesis
Such research is thus characterized by the experimenter's control over the variable under study.



Conceptual Research

Research based mainly on ideas, concepts, theories, models, frameworks, mathematical formulations or existing knowledge, rather than collecting experimental data.

- Proposing a conceptual framework for integrating LLMs into programming education.
- Developing a theoretical model for secure IoT architecture.
- Proposing a new conceptual architecture for an AI-based healthcare system.
- Developing a conceptual framework for detecting cybersecurity threats using AI.
- Designing a theoretical model for integrating RAG with an educational chatbot.

Empirical Research

Research based on observation, experiments, measurements, surveys, real-world data or other collected evidence.

- Experimentally compare YOLO models on an object-detection dataset.
- Test whether RAG reduces hallucinations in an LLM.
- Measure the performance of different database systems.
- Conduct a user study to evaluate an AI-based learning application.
- Collect programming-performance data from students and analyze the effect of AI coding tools.

Conceptual = Based mainly on ideas, theories and frameworks.

Empirical = Based on evidence collected through observation, experiments or data.

RESEARCH APPROACHES

Type of research Approaches.
Research Approaches.

Qualitative Approach

He's concerned with subjective assessment of attitude opinions and
be here

Research in such situation is a function of researchers inside and
impressions

Quantitative Approach

It involves the generation of data in quantitative form which can
be subjected to rigorous quantitative analysis in a formal and
rigid fashion



QUANTITATIVE APPROACH

GENERATING AND ANALYSIS OF DATA

INFERENTIAL APPROACHES

to form a database to infer characteristic or relationship of population. It usually means research where a sample of population is studied to determine its characteristic and it is then inferred that the population has the same characteristic

EXPERIMENTAL APPROACH

Experimental approach is characterized by much greater control over the research environment and in this case some variables are manipulated to observe their effect on the other variables.

SIMULATION APPROACHES

Simulation approach involves the construction of an artificial environment within which relevant information and data can be generated This permit and observation of the dynamic behavior of a system under control conditions

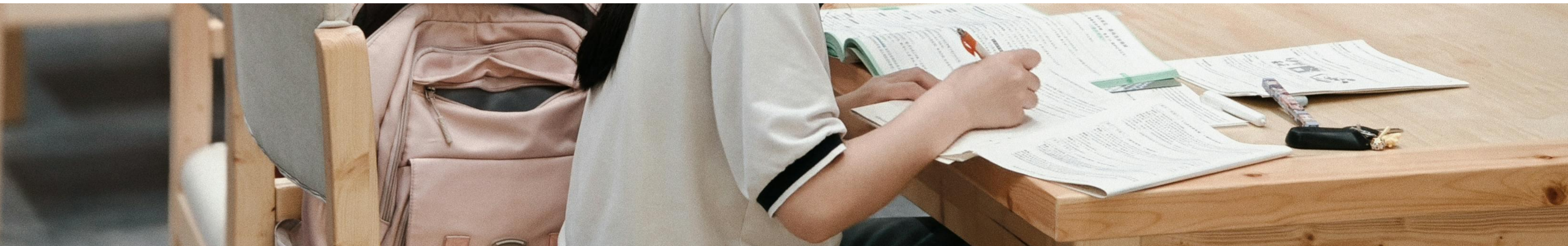
Research Methodology
Research Methodology

RESEARCH METHODS VS METHODOLOGY

Research methods or techniques, thus, refer to the methods the researchers use in performing research operations.

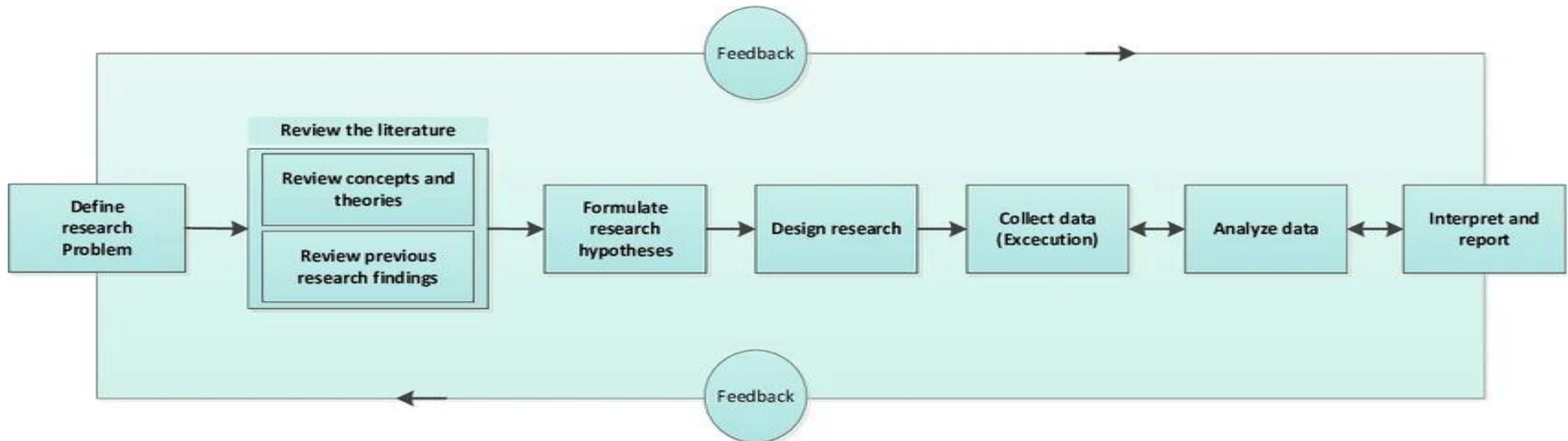
Research methodology is a way to systematically solve the research problem.

It may be understood as a science of studying how research is done scientifically. It is necessary for the researcher to know not only the research methods/techniques but also the methodology.



RESEARCH PROCESS

Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps.

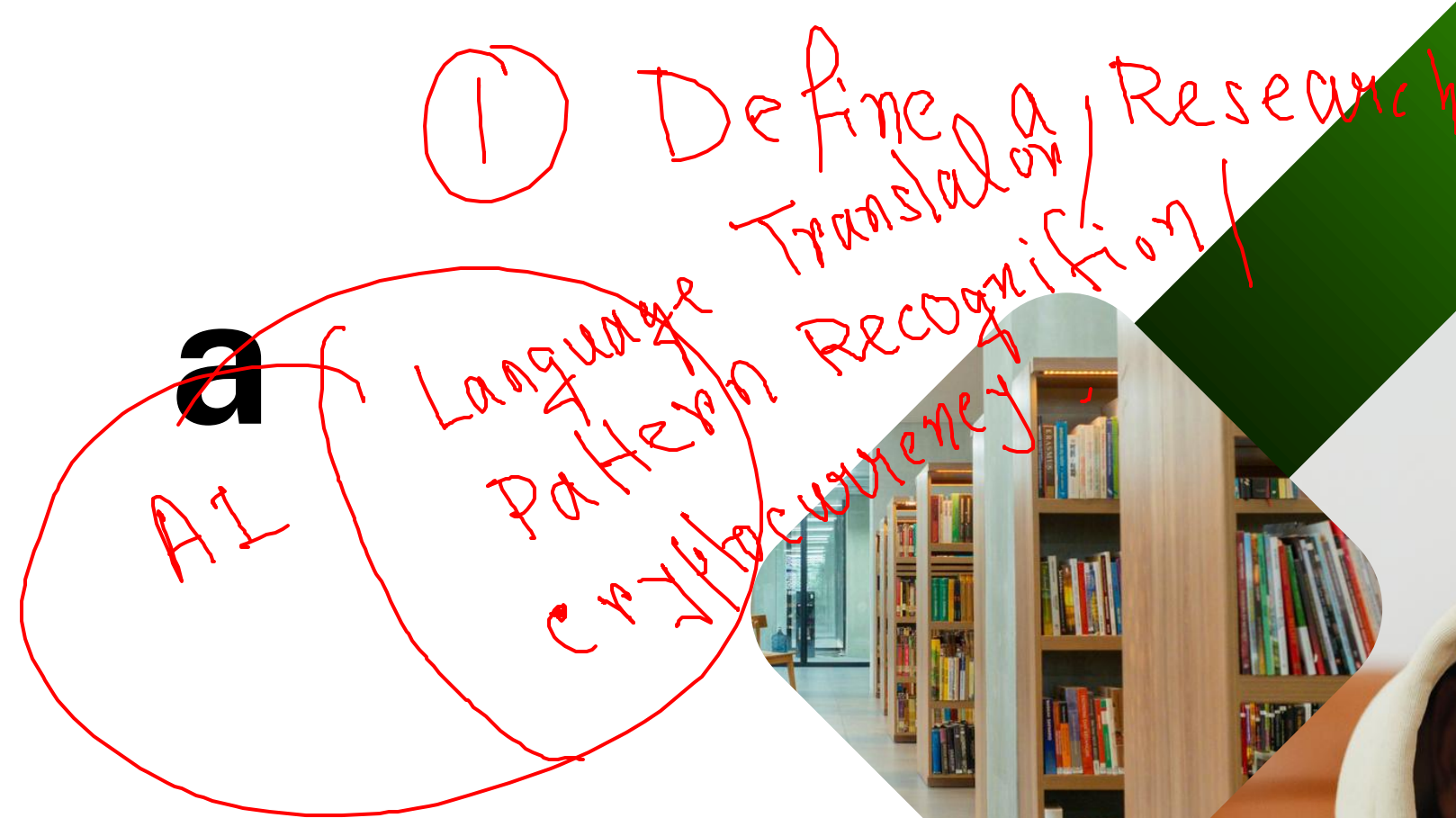


Define Research Problem

The first step is to identify something that needs to be investigated, improved, explained, or solved. Select a topic.

A good research problem should be:

- Relevant
- Clearly defined
- Feasible
- Investigable
- Related to an existing problem or knowledge gap



② Liter

Conduct a Literature Review

What has already been done by other researchers?

Students should read:

- Research papers
- Conference papers
- Books
- Review papers
- Reliable online sources



Formulating a Research Hypothesis

- ① AI / ~~to~~ Language Translation
- ② Liturg
- ③ F

Formulating a research hypothesis means making a **clear, specific, and testable prediction** about what you expect to find in your research.



Alphabet

Design Research

Word, Sentence,
Tense

2

Research design is the overall plan or blueprint of a research study. It explains how the researcher will conduct the research to answer the research questions or test the hypothesis.

1. Grammar,
Expression

Basic example
sentences need to
be converted



Collect and interpret Data

For example:

YOLO11 achieved the highest mAP and the lowest inference time among the evaluated models. This indicates that it provides a favorable balance between detection accuracy and computational efficiency for the selected low-light dataset.



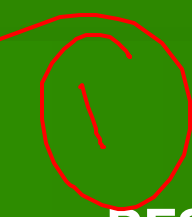
Interpret and Report

Draw the Conclusion, Identify limitation, suggest future work and communicate through research paper



CRITERIA OF GOOD RESEARCH

RESEARCH SHOULD BE
CLEARLY DEFINED



RESEARCH PROCEDURE

The research procedure used should be described in sufficient detail to permit another researcher to repeat the research for further advancement, keeping the continuity of what has already been attained.

The procedural design of the research should be carefully planned to yield results that are as objective as possible.

RESEARCH REPORT

The researcher should report with complete frankness, flaws in procedural design and estimate their effects upon the findings.

BASED ON RESEARCH DATA

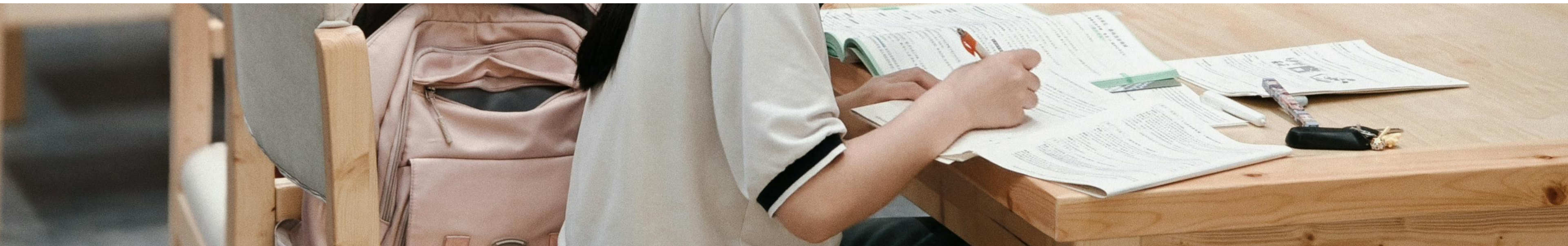
The analysis of data should be sufficiently adequate to reveal its significance and the methods of analysis used should be appropriate. The validity and reliability of the data should be checked carefully.

Conclusions should be confined to those justified by the data of the research and limited to those for which the data provide an adequate basis.

WE HAVE LEARNED IN THIS MODULE

Topics of M1

- What is research?
- Difference between project and research
- Objective of research
- Type of Research
- Research Process
- Criteria of Good research



TOPIC OF NEXT LECTURE

LITERATURE REVIEW & RESEARCH DESIGN

Research design is the overall plan or blueprint of a research study. It describes how the researcher will conduct the study to answer the research questions or test the hypothesis.



THANK YOU

I APPRECIATE YOUR TIME AND
ATTENTION TO OUR MY LECTURE.

Presented by **Dr. Tumpa Banerjee**

📞 +9883057767

🌐 www.github.com/TumpaBanerjee

✉️ tmp.manna@gmail.com



BORCELLE

